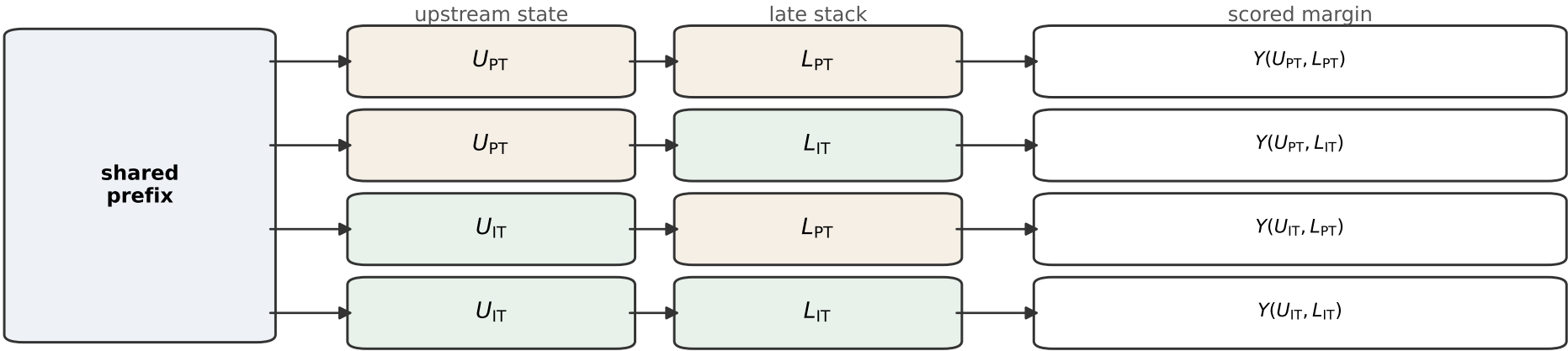


A. Four hybrid passes at the first divergence

Same raw prompt and generated prefix. Readout: $Y = \text{logit}(t_{\text{IT}}) - \text{logit}(t_{\text{PT}})$.



$$\text{interaction} = [Y(U_{\text{IT}}, L_{\text{IT}}) - Y(U_{\text{IT}}, L_{\text{PT}})] - [Y(U_{\text{PT}}, L_{\text{IT}}) - Y(U_{\text{PT}}, L_{\text{PT}})]$$

Interpretation: how much more the IT late stack helps when it reads an IT-shaped upstream state.

B. Concrete divergent-token examples

Model	Prompt fragment	pos.	PT token	IT token
Mistral 7B	Question: What is the difference between a podcast and...	4	'radio'	'digital'
OLMo 2 7B	Question: What causes the seasons? Answer:	3	' axis'	' tilt'
Qwen 3 4B	Write a startup pitch for a time capsule service. The...	5	'1'	'3'
Llama 3.1 8B	Question: My landlord is refusing to fix the heating....	13	' legal'	' several'

Examples are illustrative; all statistics use the full first-divergence support.